Remove Watermark Wondershare PDFelement

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Advanced Distance Vector
EIGRP

The router exchange routing table (It is Dynamic) with direct connected routers (It is D.V)
within the same AS (It is IGP) It is faster than any routing protocol because it use DUAL
Diffusion update algorithm and EIGRP Terms

1-Topology table

It contains all possible pathes for specific destination network address (The route with different cost)

R#sh ip eigrp topology
R#sh ip eigrp topology all-link

2-Routing table

It contains the best path foe each destination network address (The route with the lowest cost)

R#sh ip route

3-Neighbor table

It contains all EIGRP neighbor

R#sh ip eigrp neighbor

4-Successor Route

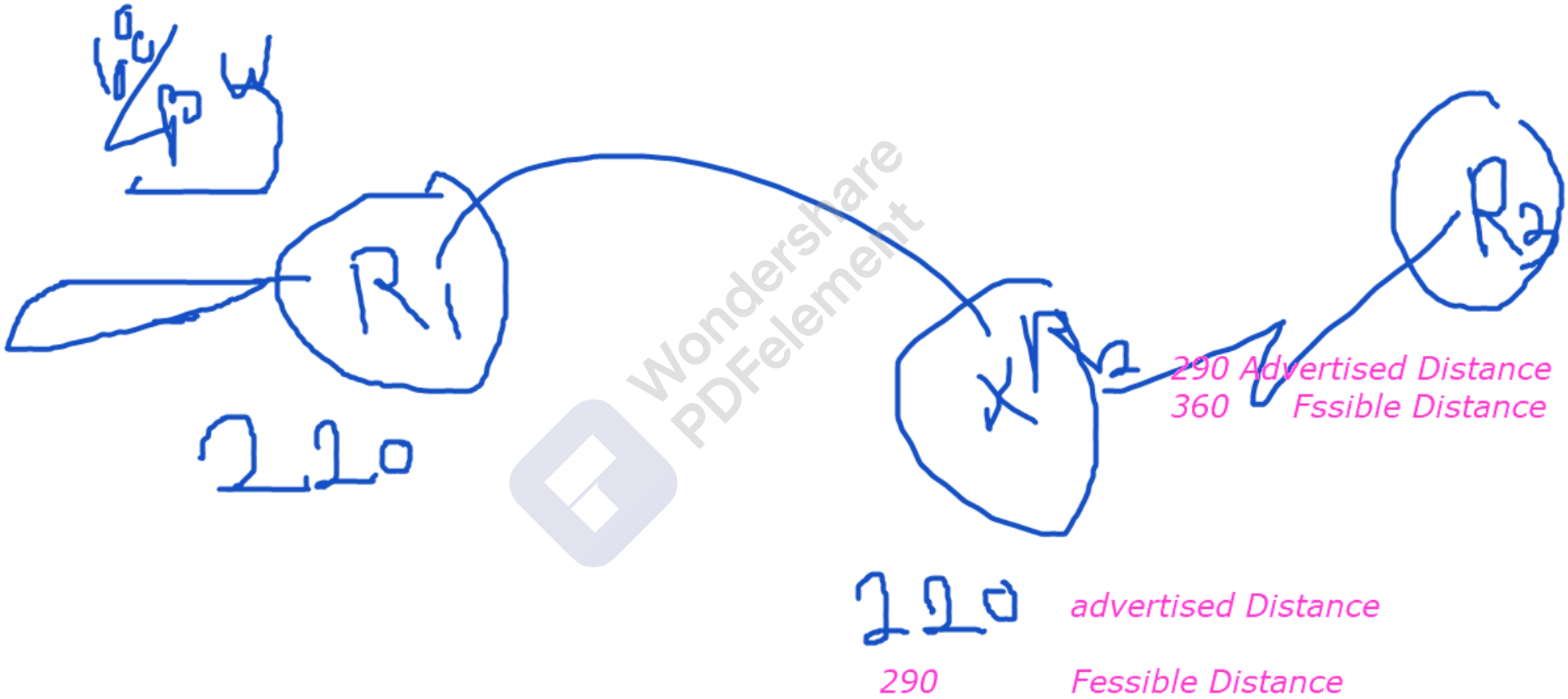
It is primary route . It is created at routing table and topology table

5-Fessible successor

It is a backup route . it is created at toplogy table only

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EIGRP configuration

R1(config)#Router eigrp AS-number

AS number is a number from 1 to 65535 which must be unique for all EIGRP router at the same AS

R1(config)#router eigrp 1
R1(config-router)#network 192.168.1.0
R1(config-router)#network 192.168.10.0



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EIGRP processing

1-At startup

A)neighbor discovery

- 1- The same AS
- 2-The same net -ID
- 3-The same K value equation
- 4-The Authentication method

B)Database exchange

C)Build the routing table

2-At convergence

Each EIGRP router will send hello packet to check that the connection a life

- At fast speed link which is link speed more than or equal 1.54Mbps , the hello will be sent each 5 sec and holding each 15 sec
- At low speed link which is the link less than 1.54 Mbps , the hello will be sent each 60 sec and hold down 180

3-At updates

At change , the router will send update packet

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EIGRP packet

1-hello packet
which is sending from source IP with destination 224.0.0.10 including (AS#, K value , Net ID , Authentication method) . this hello attribute must be the same for all EIGRP neighbor , It would be 5 sec and holding is 15 sec at high speed link (more than or equal 1.54Mbps) it is 60 sec and holding is 180 sec at slow wan link (less than 1.54 Mbps)
This hello would be set at startup to discover neighbor and at convergence to check the neighbor connectivity

2-Update packet with interface source ip and destination 224.0.0.10

3-Ack packet which is sent as unicast

4-Query Packet

5-Reply packet

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EIGRP Cost

EIGRP Cost =256 * $\left(K_1 * BW + \frac{K_2 * BW}{256 - Load} + K_3 * Delay \right) * \frac{K_5}{Reliability + K_4}$

$BW = \frac{10^7}{Bw(bps)}$ | Delay in 10 of n.s

For simplification (By default k1 , K3 is 1 and K2, K4 and K5 are 0)

EIGRP Cost = 256 (BW +Delay)

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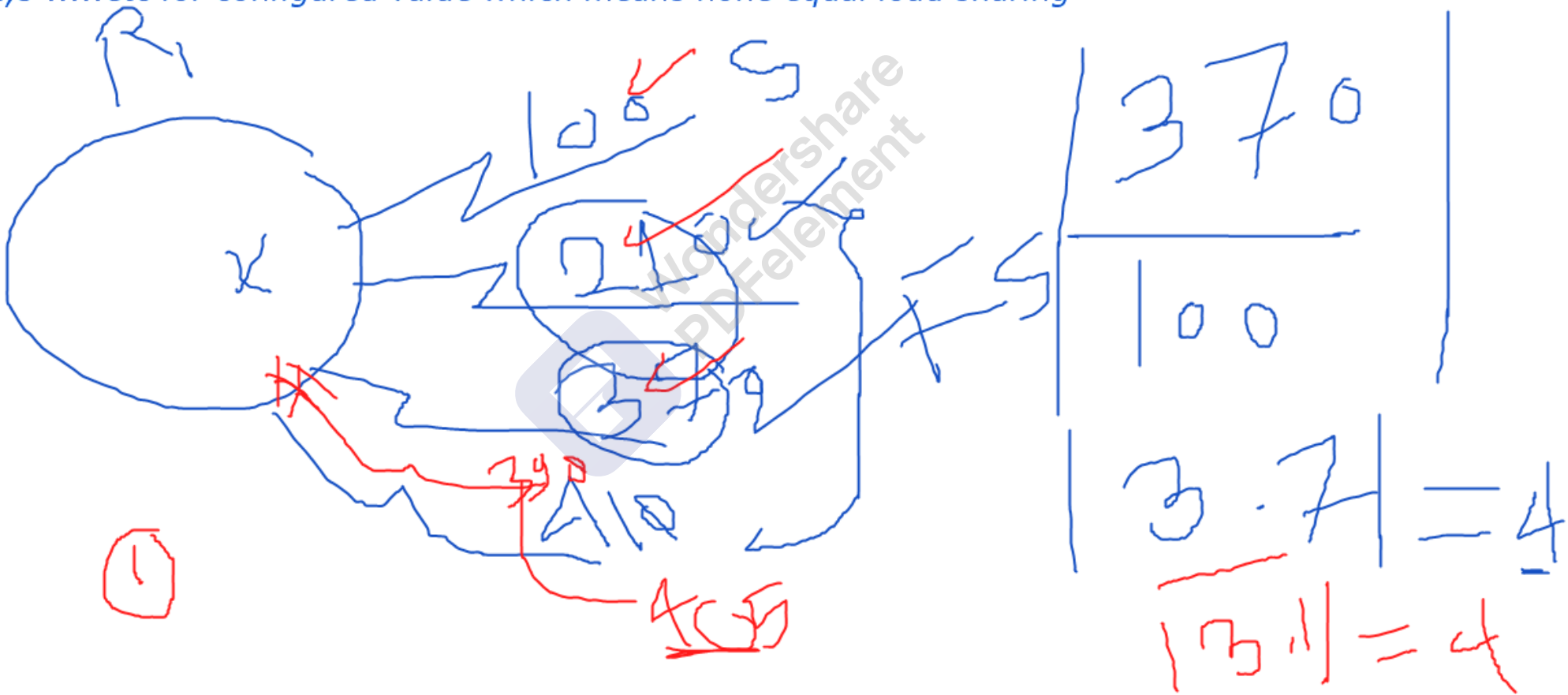
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EIGRP None Equal Loadsharing

Variance

It is a multiplier value from 0 to 128 . by default the variance is 1 which means equal load sahring

It is 2,3 .....etc for configured value which means none equal load sharing



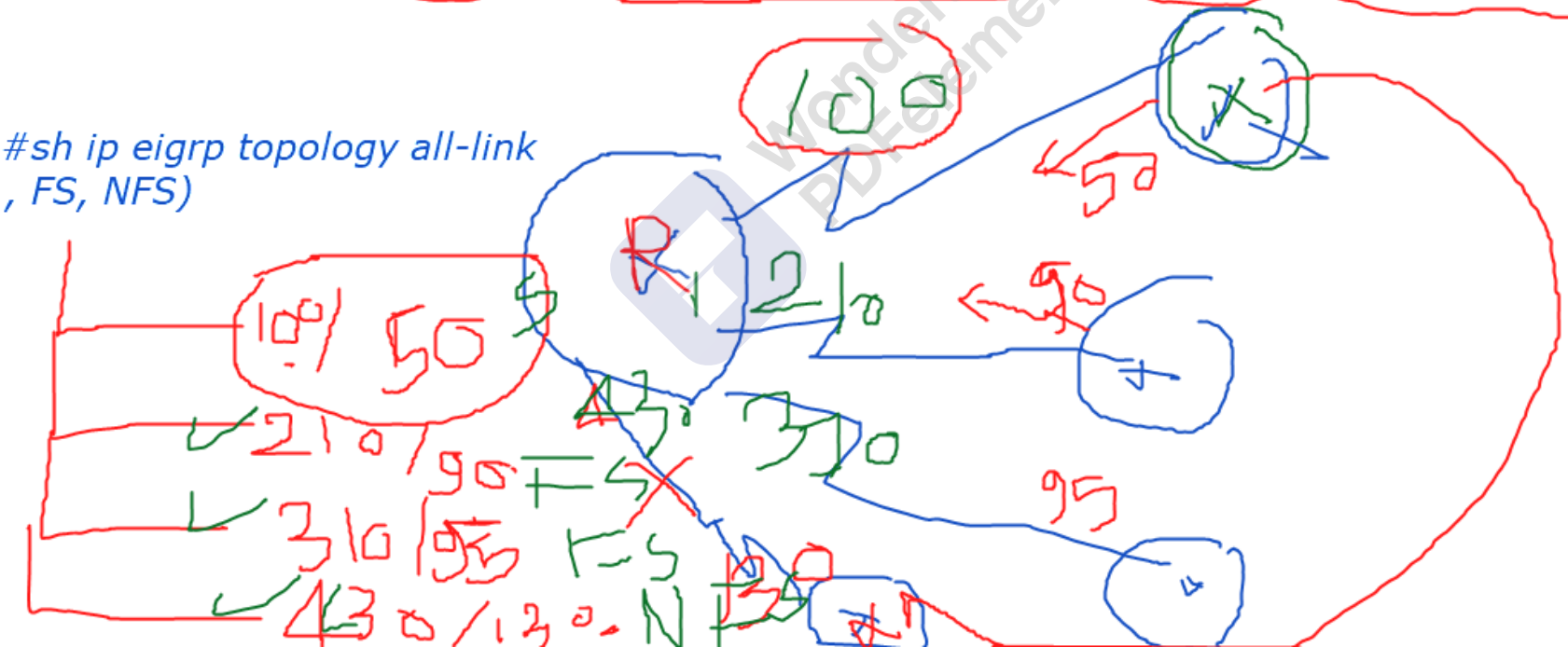
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EIGRP fessibility Condition

It is the condition which must be ocuured to store a route as a fessible successor route

Fessible Distance (Successor) must be greater than the advertised distance (fessible successor)

R1#sh ip eigrp topology all-link
(S , FS, NFS)



EIGRP Properties

1-It is a cisco protocol .

2-It uses the advanced Distance vector to build the routing table

3-AD value is 90 for Internal route D and 170 for external route .

4-Mertric is equation and BW , Delay , Load and Reliability .

5-Max hop count is 100 by default and 224 by configuration .

6-It is classless routing protocol (it sends the mask at its updates) .

7-It supports equal load sharing by default (Variance is 1) . It supports the none equal load sharing (varaiance is 2, 3,.....128)

8-It is supports multiple network layer protocol (IP, IPX , Apple talk)

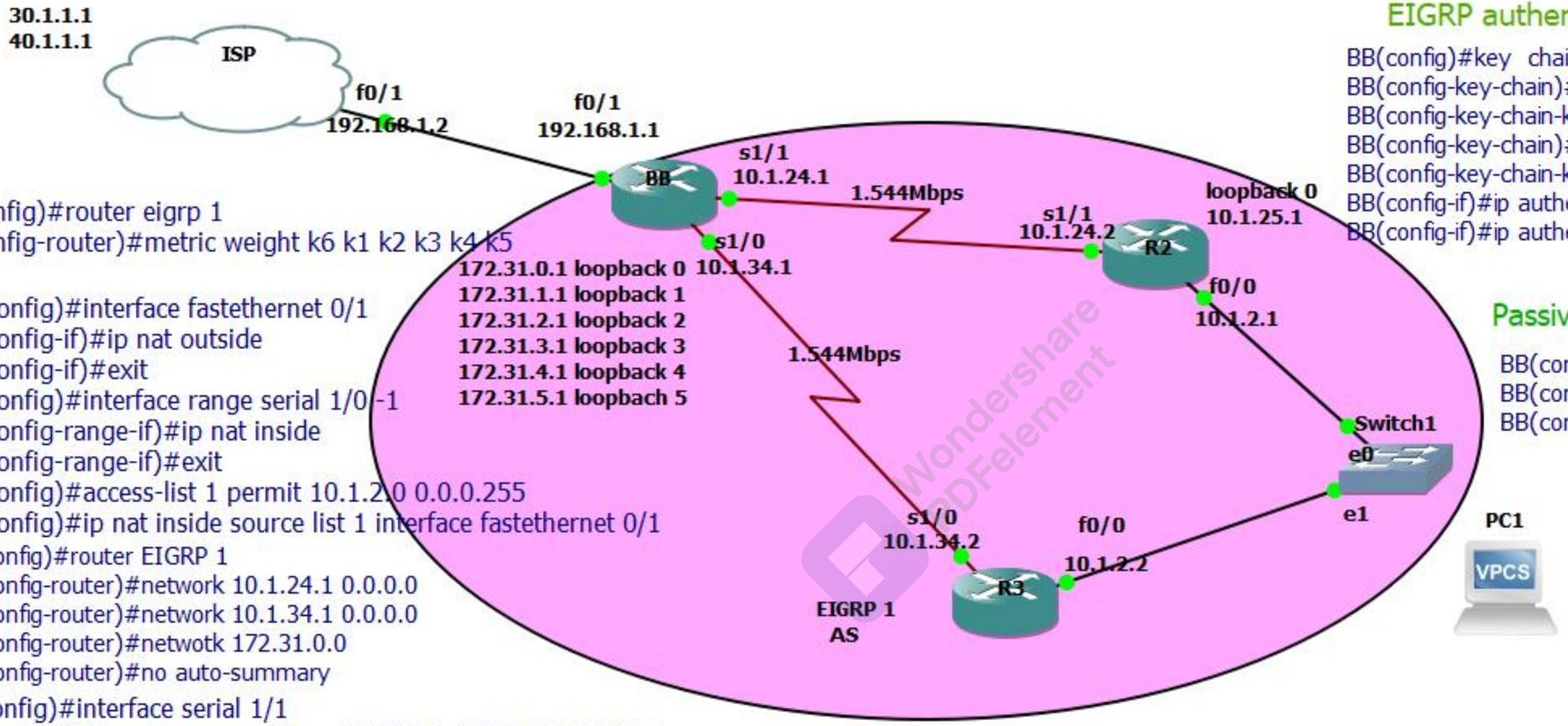
9-The updating address is 224.0.0.10

10-Symbol at Routing table is D for internal route and DEX for external Route

11-It is faster than any routing protocol because it uses DUAL algorithm

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R2(config)#router eigrp 1
R2(config-router)#metric weight k6 k1 k2 k3 k4 k5

BB(config)#interface fastethernet 0/1
BB(config-if)#ip nat outside
BB(config-if)#exit
BB(config)#interface range serial 1/0 -1
BB(config-range-if)#ip nat inside
BB(config-range-if)#exit
BB(config)#access-list 1 permit 10.1.2.0 0.0.0.255
BB(config)#ip nat inside source list 1 interface fastethernet 0/1

BB(config)#router EIGRP 1
BB(config-router)#network 10.1.24.1 0.0.0.0
BB(config-router)#network 10.1.34.1 0.0.0.0
BB(config-router)#netwotk 172.31.0.0
BB(config-router)#no auto-summary

BB(config)#interface serial 1/1
BB(config-if)#ip summary-address EIGRP 1 172.31.255.248.0
BB(config)#interface serial 1/0
BB(config-if)#ip summary-address EIGRP 1 172.31.0.0 255.255.248.0

BB(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.2
BB(config)#router eigrp 1
BB(config -router)#redistribute static

EIGRP authentication

BB(config)#key chain khaled_key
BB(config-key-chain)#key 1
BB(config-key-chain-key)#key-string gate
BB(config-key-chain)#key 2
BB(config-key-chain-key)#key-string home
BB(config-if)#ip authentication mode eigrp 1 md5
BB(config-if)#ip authentication key-chain eigrp 1 khaled_key

Passive interface

BB(config)#router eigrp 1
BB(config-router)#passive-interface default
BB(config-router)#no passive interface serial 1/1

BB(config-router)#eigrp stub connected